ERC Consolidator Grant 2024
Research proposal [Part B1]

Enabling Global-Scale Mine-Level Indicators from Satellite Observations

MINE-THE-GAP

Cover Page:
- Name of the Principal Investigator (PI): **Dr. Victor Maus**
- Name of the PI's host institution for the project: **Vienna University of Economics and Business (WU)**
- Proposal duration in months: **60 Months**

Proposal summary

The global surge in mining, driven by decarbonisation and digitalisation, has uncovered a critical gap in our understanding of its environmental and social impacts and risks. Despite the seemingly abundant knowledge about mining operations, they remain largely unmeasured, even within the European Union, posing a challenge to sustainable development efforts. Essential indicators, such as geographical locations, waste generation, and details of extracted minerals and commodities, are surprisingly scant worldwide. Satellite images and artificial intelligence algorithms are prominent technologies to fill this knowledge gap. However, scaling the mapping of mining indicators to the global level remains a challenge due to the small size and low quality of training sets. The MINE-THE-GAP project will lay the foundations of an innovative methodology to scale up the production of mine-level indicators. Methodologically, the project presents a breakthrough by integrating high-precision data products from various satellite sources to substantially enrich training sets that will enable the creation of scalable artificial intelligence models to effectively map mine land use, waste generation, and types of extracted minerals and commodities. This approach will allow the production of mine-level indicators that are timely, locally relevant, and globally reliable, and ensure the continuous improvement of models beyond the project's timeline as new high-precision data becomes available. For the first time, MINE-THE-GAP will unlock new avenues in sustainability research, overcoming previous obstacles due to the lack of essential geoinformation on mining. The project will create the basis for scalable mining site monitoring, essential for enforcing European supply chain transparency laws and achieving the United Nations' Sustainable Development Goals.

Section a: Extended Synopsis of the scientific proposal

Background and motivation

The escalating demand for mineral resources to achieve a decarbonised and digital society[1–9] has exposed a **critical knowledge gap on global mining**. Despite the pivotal role of the mining industry in sustainable development, understanding its impacts and risks globally remains a major challenge due to the **lack of spatial-temporal data**[10,11], even within the European Union (EU)[12]. Currently, information on individual mines, when it exists, is scattered and challenging to find and compile[12–15]. Mining companies provide primarily aggregated data at the corporate level instead of individual operations[11,16,17]. Furthermore, reports are mostly incomplete, as it is voluntary to report indicators necessary for sustainability assessments, such as **geolocation of tailings dams, waste volumes, area, and characteristics of waste materials**.

Recent publications, including my own work and that of others[13,14,18–20], highlight significant discrepancies and gaps between available data sets and actual on-ground realities. Even proprietary data, despite being behind substantial paywalls, are not immune to these shortcomings. For instance, 56 % of mining areas identifiable in satellite imagery lack corresponding production data in the S&P Capital IQ Pro database (formerly the S&P Global Market Intelligence SNL Metals and Mining Database)[21]. The map below illustrates the spatial distribution of these data gaps, revealing that they are indiscriminate—occurring across various commodities and locations. Moreover, the gaps in the map pertain only to production data; the absence of information on other critical factors, such as waste generation, is even more pronounced[14]. Despite their limitations, these incomplete databases are the only source of data and are used in diverse global and large-scale mining sector assessments[22–26]. This reliance can result in incorrect conclusions with high uncertainty and the potential for misguided policy.

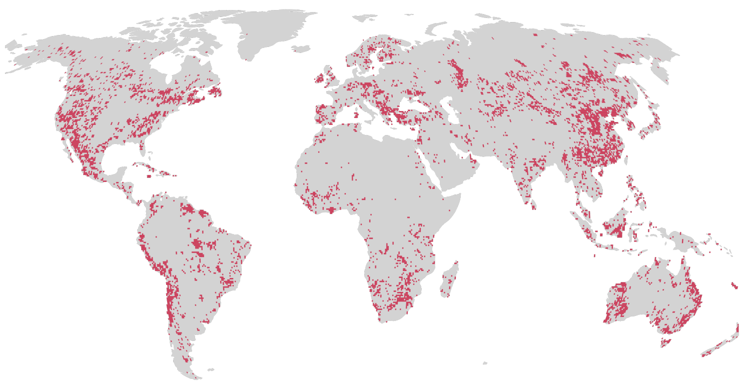

Figure B1.1: Mining regions not covered by the production data in the S&P database. Based on a lack of intersection between mine areas identified in satellite imagery and 10km buffers from production locations in the S&P database. The map is projected to the Robinson map projection and sampled to 50 x 50 km grid cells to facilitate visualisation. Source: Maus (2023)[21].

The increasing availability of **satellite Earth observation (EO)** data is key in bridging knowledge gaps in global mining. Consolidated remote sensing methods can use EO to create detailed mineral maps[27,28] and accurate terrain change measurements[29,30], that for instance, can offer insights into mining waste flows[19]. However, these methods rely on specific satellite sensors, limited by their lack of global coverage or short lifespans, restricting their global-scale application in dynamic sectors like mining. In contrast, data-driven **artificial intelligence (AI)**[*] methods using satellite sensors with global, long-term coverage show promise in generating relevant mining variables, like infrastructure mapping[31–41] and height estimation[42,43]. Yet, scaling up AI models to larger areas requires extensive training sets, currently unavailable for mining sites[44].

*MINE-THE-GAP will use AI algorithms exclusively to produce information about mining sites from publicly available EO data sources and, therefore, does not raise any ethical issues.

The MINE-THE-GAP proposes an innovative research programme that will **develop the methodological foundation to enable global mining spatial-temporal information**, revealing where and when mining occurs, how much material is displaced, and what commodities and minerals are related to individual mining operations. MINE-THE-GAP will address the following overarching research question: **How can we produce timely, locally relevant, and globally reliable mine-level indicators?** Methodologically, the project presents a breakthrough by integrating high-precision data products from various satellite sources, including hyperspectral, laser, and radar, to substantially enrich training sets that will enable the creation of scalable AI models to map mining indicators effectively. This approach will, for the first time, produce global-scale mine-level indicators to support risks and impact assessments of mines and provide the basis for more comprehensive global sustainability analyses of mining regions.

a.1 Objectives

MINE-THE-GAP's aim is to develop and apply methods to produce timely basic indicators at the level of individual mining sites that are locally relevant and at the same time spatially and temporally consistent to allow for global and regional impact and risk assessments.

The following five specific objectives are defined, each clearly linked to the project's work packages:

Objective 1 Construct a global-scale AI-ready database covering more than 80,000 polygons, including detailed semantic labelling of mining infrastructure types such as open pits, overburden piles, waste rock piles, and tailings dams, terrain heights within mines and minerals and commodities information (WP 1, WP 2, and WP 3).

Objective 2 Develop accurate and scalable models to automatically detect and map mining facilities such as open pits, overburden piles, waste rock piles, and tailings dams (WP 1).

Objective 3 Develop accurate and scalable models to estimate waste generation indicators based on terrain changes within mining infrastructure types such as open pits, overburden piles, waste rock piles, and tailings dams (WP 2).

Objective 4 Develop accurate and scalable models to detect commodities and minerals in individual mining facilities (WP 3).

Objective 5 Integrate and validate the developed models into a framework to produce basic spatial-temporal indicators for individual mining facilities (WP 4).

Originality and impact

The MINE-THE-GAP project will deliver a **scientific breakthrough** by designing a methodology capable of producing timely, locally relevant, and reliable global-scale mine-level indicators. Its **interdisciplinary nature** positions the project to critically contribute to the advancement of the research frontier in several disciplines. In geoinformatics and remote sensing, the project will enhance understanding of how to strategically combine various satellite data sources, addressing a key challenge in scaling AI training for global EO applications. The approach introduced by the project will facilitate the creation of AI models that are capable of continuous improvement, harnessing the potential of highly accurate data from both current and future satellite missions. Moreover, we will produce the largest openly available AI-ready database for mining sites that will serve as a benchmark to develop and test new AI algorithms.

MINE-THE-GAP holds **disruptive potential** in sustainability research by significantly expanding the methodologies for studying the global mining sector using EO images, independent of company reports. This project will, for the first time, enable the identification of what is being mined, the volume of displaced materials, and the location and timing of the activities. These will overcome the existing limitations of incomplete and low-quality spatial data on mining typically extracted from reports, providing a new set of essential indicators from which further sustainability assessments can expand and build on. The new mining indicators can be used to improve total material footprint accounts[45,46], evaluate circular economy initiatives[14,15], assess the impacts of global supply chains[47,48], and study mineral resource governance[11] and mining reclamation[49]. Georeferenced mining infrastructures, such as tailings dams, combined with waste volume and mineral indicators, will substantially enhance the robustness of comprehensive mining pollution and disaster risk assessments[23,50]. The new spatial indicators can increase the scope of variables to assess the impact of mining operations in the Earth System[16]. Further, the capability to provide timely information on mine expansion will impact global land use change assessments and monitoring, as well as forest and biodiversity loss studies[51].

MINE-THE-GAP stands to make a significant impact on transparency in the global mining sector, addressing an **urgent theme** with substantial potential for impact beyond academia, including policy, industry, finance and civil society. The project outcomes will enhance the availability of crucial information that can potentially support local communities, NGOs, and other actors in making informed decisions and holding the mining industry accountable. The outcomes of the project have the potential to enhance evaluations of corporate environmental impacts and guide decisions regarding the allocation of public and private investments. The project can support key international agreements such as the Paris Agreement, UN Sustainable Development Goals, and the Kunming-Montreal Global Biodiversity Framework by offering new spatial data that can be linked, for example, to carbon emissions, biodiversity loss, ecosystem restoration potential and indigenous land rights. MINE-THE-GAP will enable insights to support the European Green Deal, enhancing knowledge on mining's environmental impact within and outside the EU, which are also relevant to new EU policies like the Corporate Sustainability Reporting Directive and the Corporate Sustainability Due Diligence Directive.

To ensure that the outcomes of the project reach a wider audience, results will be disseminated through the project's website, high-impact journals, social media, and conferences, and all models, codes, and data will be freely available (e.g., on Github at `www.github.com/mine-the-gap`). My existing networks in academia and the private sector will be regularly updated on project progress, ensuring that MINE-THE-GAP's research outcomes are enduring and continue to offer **valuable insights into global mining well beyond the project's conclusion**.

Methods and implementation

MINE-THE-GAP will be implemented in four work packages. WPs 1-3 are designed to deal with one dimension of mining indicators each, as illustrated in Figure B1.2. WP 4 will integrate and validate key innovations realised in WP 1, WP 2, and WP 3 into a framework to produce mine-level indicators.

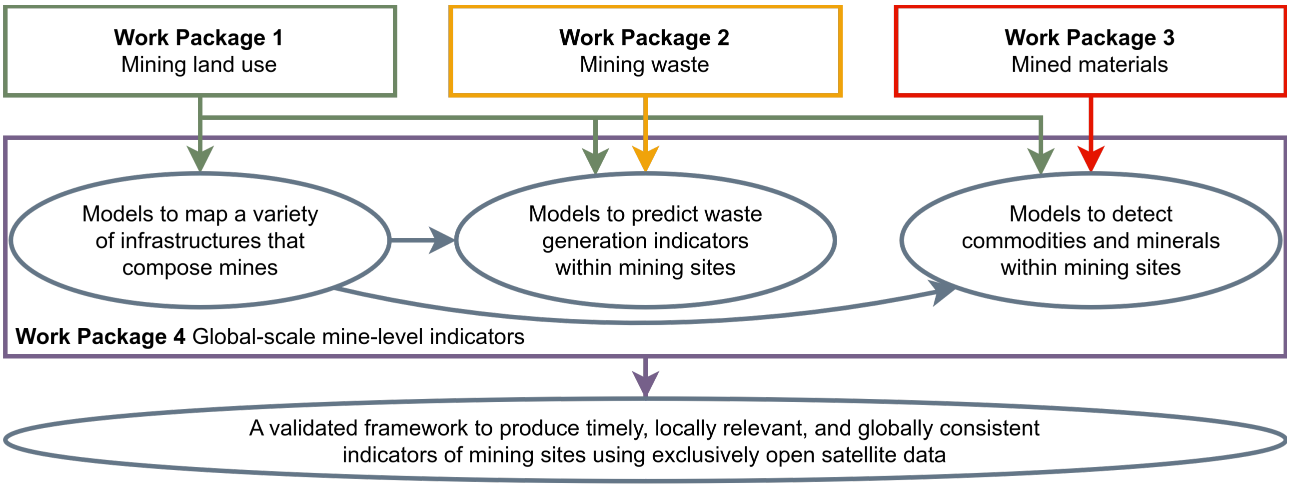

Figure B1.2: Structure of the MINE-THE-GAP project.

WP 1: Mining land use (years 1-3)

WP 1 addresses Objective 1 and Objective 2. It will create a large AI-ready database to train models for semantic segmentation of mining areas, starting from 81,562 non-overlapping polygons from my team's previous mapping initiatives[18,52] and colleagues[53]. We will re-interpret these polygons to add information on mining infrastructure types, such as open pits, overburden piles, waste rock piles, and tailings dams. This will require a substantial amount of work, which we will expedite by using an assisted-segmentation approach[54] integrated into our open-source tool, that has been designed to map mining areas[18,55]. The labelled images will consist of radar and optical images from Sentinel-1 and Sentinel-2, chosen for their proven synergy in enhancing predictions of heights[42,43], land cover types[56], and mining tailings[40], along with their long-term global coverage. As an output, the new large AI-ready database will substantially increase the training set size and reduce spatial bias in AI model training, enabling the development of reliable models to map mining land use globally.

WP 2: Mining waste (years 1-3)

WP 2's main goal is to address the modelling of indicators of mining waste generation, MINE-THE-GAP's Objective 3. We envision developing waste indicators based on the mining infrastructure types and their height change over time. Therefore, WP 2's first task is to enhance our AI-ready database with height information, contributing to Objective 1. Terrain elevation data from various sources will be compiled, particularly the accurate 3D data from NASA's ICESat-2[57] and GEDI[58] Light Detection and Ranging (LIDAR) missions. Additional high-resolution Digital Elevation Models (DEMs) from TanDEM-X and TerraSAR-X satellite images will be generated for selected mining locations[59–61]. In addition, the open TanDEM-X 30m Edited DEM product[62–64] alongside country-specific open DEMs like the USGS's 3D Elevation Program[65] will also compose valuable assets to assemble a large training set. The comprehensive training set on terrain heights will support the development of scalable models to map terrain changes[42,43], providing essential data for estimating the volume of overburden materials, tailings, and overall volumetric changes within mines.

WP 3: Mined materials (years 3-5)

This work package develops models to detect commodities and minerals within mines, addressing Objective 3. It will also complete Objective 1 by producing an extensive global training set of minerals and commodities that will be integrated into a final version of the AI-ready database. Two sources of data will be integrated. The first is mineral information derived from hyperspectral sensors[27,66], specifically images acquired by the EnMAP[28], Prisma[67], EO-1 - Hyperion[68] missions. This will be supplemented with information from company reports and proprietary datasets[13,69] to enhance the AI-ready database. This large data set with information on commodities and minerals will allow us to explore various AI approaches to detect minerals and commodities within mines globally.

WP 4: Global-scale mine-level indicators (years 3.5-5)

WP 4 addresses MINE-THE-GAP's Objective 5. It will focus on integrating and validating the models developed in the previous work packages into a framework to produce mining indicators. It goes beyond typical AI model development by not only generating standard performance metrics through cross-validation[70] but also validating our models against a range of independent data sources. This comprehensive validation will use data from both open and proprietary sources, with our primary source being my team's own compilation of 1,171 mines extracted from company reports detailing the extraction of 80 materials from 2000 to 2021[13,71]. Additional validation will come from open sources like the Tailings Database and Portal[72], Global Coal Mine Tracker[73], the Mineral Resources Online Spatial Data[74], data from several scientific publications[40,75–79], as well as proprietary databases like S&P[69]. Key resources for validating terrain changes include the TanDEM-X 30m DEM Change data product[29,30] and for land use and geolocation validation, we will use the high-resolution imagery from

NICFI[80], Google Maps, and Open Street Maps[81]. Furthermore, we also plan country-level assessments using mineral extraction and waste flow statistics[82–85] to compare with official data and identify discrepancies. With this approach and support from several data sources, we will be able to identify the strengths and weaknesses of the project approach and guide further developments.

Feasibility

MINE-THE-GAP is an ambitious project that **addresses a major knowledge gap in global mining in an original and interdisciplinary manner** by integrating various methods and data sources that have not been linked before and, therefore, can pose several risks. However, pre-existing conditions and complementary measures will minimise potential threats to the project implementation and ensure the achievement of high gains with limited risks. *First* **I have a solid scientific foundation in fields that are essential for the success of MINE-THE-GAP**, including geoinformatics and big-EO data analytics and machine learning combined with knowledge and experience in global mining assessment. To complement my research expertise, **I have carefully defined five profiles of team members that will ensure a successful implementation** of the planned research. Specifically, I will recruit a postdoctoral researcher expert in radar satellite imagery for 3D modelling to lead WP 2, a postdoctoral researcher with expertise in spaceborne imaging spectroscopy applied to mineral mapping to lead WP 3 and three PhD students with a solid background in AI/ML applied to EO data, one for each the three first work packages.

Second **several building blocks that are critical to the planned research are already in place**, with promising results. These refer mainly to recently compiled large-scale mining datasets[13,18,52], which I have led and contributed to. The new databases proposed in MINE-THE-GAP will be challenging to achieve, but their development will count on my extensive experience collecting mining data, ensuring that we will produce sufficiently large sets to train AI models. *Third* the access and processing of a large volume of satellite images are critical for implementing MINE-THE-GAP. To address the risk of not having access to enough data, **I have planned mainly images from Sentinels, which have global continuous coverage and are planned for long-term open data provision**. In addition, Sentinel images are available directly from several cloud processing facilities, e.g. Google Earth Engine, Amazon Web Services, and Planetary Computer. Other satellite images with limited usage quotas are planned only in small quantities to extend the reference database.

Confirmed collaboration of researchers from globally recognised institutions significantly reinforces the feasibility and importance of this project. In WP 1, we will exchange experience and collaborate on image segmentation and mine mapping with Prof. Marius Apple from Bochum University of Applied Sciences (Germany), Prof. Alex Lechner at Monash University (Indonesia), and Prof. Sidnei Luís Bohn Gass at the Federal University of Pampa (Brazil). Dr. Marie Lachaise from the German Aerospace Centre (Germany) will contribute her expertise on terrain modelling to WP 2. For WP 3, Prof. Martin Herold at the GFZ German Research Centre for Geosciences has confirmed his collaboration, particularly on hyperspectral data analysis. Furthermore, the development and validation of global-scale mine-level indicators in WP 4 is of interest to several key researchers on mining and sustainability worldwide, including Dr Tim Werner from the University of Melbourne (Australia), Prof. Stefan Gljum at the Vienna University of Economics and Business (Austria), Dr Ian McCallum from the International Institute for Applied Systems Analysis (Austria), Prof. Michael Tost at the University of Leoben (Austria), and Prof. Kai Qin from the China University of Mining and Technology (Xuzhou, China). These confirmed **collaborations will enrich the project's intellectual environment and facilitate exchanges**, thereby enhancing the project's implementation and reducing risks.

The combination of existing conditions and supplementary strategies will ensure **controlled risk and high gains**, leading MINE-THE-GAP to a scientific breakthrough in enabling global-scale mine-level indicators, which holds the potential for generalisation, setting the stage for scalable models in various research fields.

Section b: Curriculum vitae and Track Record

PERSONAL DETAILS

Last name, First name:	Maus, Victor (Alternative name: Wegner Maus, Victor)
Researcher unique identifiers:	ORCID: 0000-0002-7385-4723 Scopus author ID: 56246500300 Researcher ID: D-9547-2013 Google Scholar: wN2LseQAAAAJ&hl
URL for web site:	https://research.wu.ac.at/en/persons/victor-wegner-maus-7

- **Education and key qualifications**

29/Apr/2016	PhD Earth System Science Center, National Institute for Space Research (INPE), Brazil Prof. Dr. Gilberto Câmara
2011	Master of Sciences Computational Modeling, Federal University of Juiz de Fora (UFJF), Brazil, Brazil
2009	Bachelor of Sciences Environmental Engineering, Franciscan University (UFN), Brazil

- **Current position(s)**

2018–	Senior Researcher Institute for Ecological Economics, Vienna University of Economics and Business (WU), Austria
2016–	Research Scholar Advancing Systems Analysis Program, International Institute for Applied Systems Analysis (IIASA), Austria

- **Previous position(s)**

2014–2016	Research Assistant Institute for Geoinformatics (Ifig), University of Münster (WWU), Germany
2012–2014	Assistant professor Science and Technology, Federal University of Pampa (UNIPAMPA), Brazil
2011–2012	Research Assistant Earth System Science Center, National Institute for Space Research (INPE), Brazil
2009–2011	Teaching assistant Computational Modeling, Federal University of Juiz de Fora (UFJF), Brazil

- **Supervision of students**

2018 - 2023	4 Master students and 4 Bachelor students, Ecological Economics, WU, Austria
2022 - 2023	1 Master students, Environmental Sciences, Wageningen University, Netherlands
2020 - 2021	1 Bachelor student, Survey Engineering, UNIPAMPA, Brazil
2017 - 2018	1 Master student, Computer Science, Univ. of Appl. Sci. Wiener Neustadt, Austria
2017 - 2017	2 PhD students in the Young Scientists Summer Program, IIASA, Austria

- **Institutional responsabilities**

2013–2014	Deputy Coordinator of Bachelor Program Sci. and Technology, UNIPAMPA, Brazil
2012–2014	Member of a Committee for Computer Infrastructure, UNIPAMPA, Brazil

- **Research funding**

2024–2024	EUR 23.937,00 Research grant (PI), funder: WU, Austria
2022–2023	EUR 30,000.00 Seed funding for project writing (PI), funder: WU, Austria
2021–2022	EUR 30,000.00 Research grant (PI), funder: The World Bank, USA
2020–2022	EUR 249,508.00 Research grant (Co-applicant), funder: FFG, Austria
2019–2020	EUR 20,000.00 Research grant (PI), funder: WU, Austria

RESEARCH ACHIEVEMENTS AND PEER RECOGNITION

Research achievements

As one of my most significant recent **scientific achievements**, I led research on "mapping global mining areas using Earth observation (EO) data" published in the journal Nature Scientific Data in 2020 (1) and available from an open data repository (2). This research revealed for the first time the extent of the land footprint of the global mining sector, which has had a high impact on academia (cited more than 153 times in Google Scholar and 98 times in Scopus) and beyond. For instance, it has been integrated into databases (e.g. Statista Database company[86]) and featured by media outlets (Mine Magazine[87], Mineral-choices.com[88], Mining-technology.com[89], and BBC Future Planet[90]). This positive feedback motivated me to lead an update on global mining land use published in 2022 (3), which is also openly available (4). I have also addressed various interdisciplinary research questions. I co-lead a study on mining and deforestation (5) and contributed to define actions to align global climate and conservation goals due to global mining's threats to biodiversity (6), both articles published by the Proceedings of the National Academy of Sciences (PNAS). In addition, I conceptualised and implemented research that revealed the risks of coal mining in Indonesia (7) in Ambio, and quantified the growing threats of metal mining to vulnerable ecosystems (8) in Global Environmental Change, and co-designed the first prototype to measure the total impact of mining activities on the Earth system (9), published in the Journal of Cleaner Production.

Furthermore, a standout achievement in my career is the development of the Time-Weighted Dynamic Time Warping (TWDTW), a machine learning algorithm tailored for Earth observation time series classification. I designed TWDTW to mitigate the effects of small and biased training sets, thereby enabling large-scale EO applications (10). This innovation has gained significant recognition, as evidenced by its citation count – over 239 times in Google Scholar and 173 times in Scopus. I have also developed the R package dtwSat (`https://github.com/r-spatial/dtwSat`), an open-source software offering a user-friendly interface for TWDTW. This package has achieved over 45,000 downloads, demonstrating its widespread adoption in both scientific and commercial settings.

Drawing on my extensive experience in big-EO analytics and global mining, I have recognised the crucial role that EO can play in generating urgently needed spatial information about mining globally, which led me to conceive and design the project MINE-THE-GAP. My early scientific achievements and my demonstrated independent thinking put me uniquely positioned to lead MINE-THE-GAP and develop new methods to enable mine-level global-scale mining indicators.

(1) **Maus**, V., Giljum, S., Gutschlhofer, J., da Silva, D. M., Probst, M., Gass, S. L., Luckeneder, S., Lieber, M. & McCallum, I. A global-scale data set of mining areas. *Scientific Data* 7, 1–13 (2020).

(2) **Maus**, V., Giljum, S., Gutschlhofer, J., da Silva, D. M., Probst, M., Gass, S. L., Luckeneder, S. &

McCallum, I. Global-scale mining polygons Version 1. *PANGAEA* https://doi.org/10.1594/PANGAEA.910894 (2020).

(3) **Maus**, V., Giljum, S., da Silva, D. M., Gutschlhofer, J., da Rosa, R. P., Luckeneder, S., Gass, S. L. B., Lieber, M. & McCallum, I. An update on global mining land use. *Scientific Data* 9, 433 (2022).

(4) **Maus**, V., da Silva, D. M., Gutschlhofer, J., da Rosa, R., Giljum, S., Gass, S. L. B., Luckeneder, S., Lieber, M. & McCallum, I. Global-scale mining polygons Version 2. *PANGAEA* https://doi.org/10.1594/PANGAEA.942325 (2022).

(5) Giljum, S., **Maus**, V., Kuschnig, N., Luckeneder, S., Tost, M., Sonter, L. J. & Bebbington, A. J. A pantropical assessment of deforestation caused by industrial mining. *PNAS* 119 (2022).

(6) Sonter, L. J., Maron, M., Bull, J. W., Giljum, S., Luckeneder, S., **Maus**, V., McDonald-Madden, E., Northey, S. A., Sánchez, L. E., Valenta, R., Visconti, P., Werner, T. T. & Watson, J. E. M. How to fuel an energy transition with ecologically responsible mining. en. *PNAS* 120, e2307006120. (2023).

(7) Werner, T., Toumbourou, T., **Maus**, V., Lukas, M.C., Sonter, L.J., Muhdar, M., Runting, R.K. & Bebbington, A. Patterns of infringement, risk, and impact driven by coal mining permits in Indonesia. *Ambio* (2023).

(8) Luckeneder, S., Giljum, S., Schaffartzik, A., **Maus**, V. & Tost, M. Surge in global metal mining threatens vulnerable ecosystems. *Global Environmental Change* 69, 102303 (2021).

(9) Crona, B, Parlato, G., Lade, S., Fetzer, I., **Maus**, V. Going beyond carbon: An Earth system impact score to better capture corporate and investment impacts on the earth system. *Journal of Cleaner Production* 429 (2023).

(10) **Maus**, V., Camara, G., Cartaxo, R., Sanchez, A., Ramos, F. M. & De Queiroz, G. R. A Time-Weighted Dynamic Time Warping Method for Land-Use and Land-Cover Mapping. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 9, 3729–3739 (2016).

Peer recognition

My early research contributions have gained increasing international recognition, as evidenced by five examples. Most recently, I was invited by Nature Journal to author a Commentary piece on the dearth of data for global mining assessments and propose potential solutions (1). This commentary is currently in the final editing stage and is scheduled for publication in Nature's January 4 issue. In 2023, I received the Young Scientist Award from the International Society for Mine Surveying (2), recognising my significant scientific contributions to the field, particularly in assessing global mining impacts. Further highlighting these achievements, in January 2022, I was honoured as WU's Researcher of the Month (3), a series recognising the research excellence of distinguished scholars at WU. My expertise in global mining has also led to invitations to speak at the XVIII International Congress for Mine Surveying (4) on 'Challenges in Big-Earth Observation Data Analytics for Global-Scale Mining Land Use Surveying.' Additionally, I was invited to address various stakeholders and policymakers about 'Mapping Global Mining Activities Using Open Earth Observation Data' (5) at a workshop on geospatial data and illegal mining organised by the United Nations Office on Drugs and Crime (UNODC) and the UN Environment Programme (UNEP). This growing international recognition of my research program has also catalysed opportunities for collaboration with renowned institutions, which will enhance the implementation of MINE-THE-GAP.

Furthermore, my research has also taken the attention of media outlets, which interviewed me about the research on mining, including the Mine Magazine (6), BBC Future Planet (7) and Radio Österreich 1 from Austria (8). In addition, I have also provided an expert contribution to a World Wildlife Fund (WWF) report on mining and deforestation (10) and to an elicitation process on the environmental impacts of mining for sustainable finance (9). This wide coverage highlights the societal relevance of my research and sheds light on the potential high impact of MINE-THE-GAP.

(1) Nature Commentary article Working title "Half the world's mines are undocumented – for a truly green energy transition, that must change" (2024).

(2) Award – Young Scientist Award from the International Society for Mine Surveying (ISM) `https://www.victor-maus.com/assets/img/award-ism-2023.jpg` (2023).

(3) Award, Researcher of the Month, January 2022, Vienna University of Economics and Business (WU), Vienna, Austria `https://www.wu.ac.at/forschung/wu-forscherinnen/rom22/victor-wegner-maus/` (2022).

(4) Invited talk at the International Congress for Mine Surveying, Oct 26-29, 2023, Xuzhou, China. Title: Challenges in Big-Earth Observation Data Analytics for Global-Scale Mining Land Use Surveying, slides: `https://www.victor-maus.com/assets/talks/2023-10-28-ims` (2023).

(5) Invited talk at the United Nations Office on Drugs and Crime (UNODC) and UN Environment Programme (UNEP), Apr 19, 2023, Vienna, Austria. Title: Mapping global mining activities using open Earth observation data, slides: `https://www.victor-maus.com/assets/talks/2023-04-19-UN` (2023).

(6) Interview with Mine Magazine. Full disclosure: mapping mining at the global scale `https://mine.nridigital.com/mine_dec20/mining_data_satellite` (2020).

(7) Interview with BBC Future Planet. How ending mining would change the world `https://www.bbc.com/future/article/20220413-how-ending-mining-would-change-the-world` (2022).

(8) Interview with ORF Science. Bergbau braucht 100.000 Quadratkilometer Erdoberfläche `https://science.orf.at/stories/3211484/` (2022).

(9) Invited contribution to an expert elicitation process on the environmental impacts of mining for sustainable finance organised by the Stockholm Resilience Centre at Stockholm University and the Global Economic Dynamics and the Biosphere programme at the Swedish Royal Academy of Sciences (2022).

(10) Expert contribution to the WWF report Extracted forests: unearthing the role of mining-related deforestation as a driver of global deforestation `https://www.wwf.de/fileadmin/fm-wwf/Publikationen-PDF/Wald/WWF-Studie-Extracted-Forests.pdf` (2023).

ADDITIONAL INFORMATION

Career breaks, diverse career paths and major life events

Jan 1, 2023 –	Part-time work at 60% of a full time employment
Mar 17, 2023 – Apr 16, 2023	Paternity leave for one month
May 1, 2021 – Aug 31, 2021	Paternity leave for four months
May 1, 2020 – May 31, 2020	Paternity leave for one month

Other contributions to the research community

(1) For more than eight years, I have contributed to developing scientific open-source software for geospatial data analysis using languages such as Fortran, C, R, and Python. As an active member of the r-spatial community (`www.github.com/r-spatial`), I have been developing and maintaining open-source tools. This includes the R packages *twdtw* (`www.github.com/vwmaus/twdtw`), *dtwSat* (`www.github.com/r-spatial/dtwSat`), and *sits* (`www.github.com/e-sensing/sits`), which together provide user-friendly, efficient, and scalable satellite time series classification for large areas. These activities highlight my commitment to open science and to developing tools and methods which can further enhance the project's implementation through community engagement to produce highly relevant outcomes.

References

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